

Brainstorm & Idea Prioritization Template

Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value

Team ID: SWTID-2026-2896

Duration: 3 Hours

Step 1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement

In ServiceNow, unauthorized users may attempt to access sensitive student records belonging to different academic branches. Without proper access control, users may view, create, modify, or delete records that they are not authorized to access.

Objective

To implement script-controlled Access Control Lists (ACLs) in ServiceNow to restrict access to student records based on the Branch field and user roles. EEE users with the required roles can access EEE records according to their permissions, while administrators retain full access.

Team Collaboration

The team discusses the security challenges associated with student records, identifies unauthorized access as the main problem, and explores role-based and field-based access control mechanisms. Team members collaborate to design and test READ, CREATE, WRITE, and DELETE ACLs.

Step 2: Brainstorm, Idea Listing and Grouping

Brainstormed Ideas

1. **Role-Based Access Control:** Assign specific roles to users to control their permissions.
2. **Field-Based Access Control:** Restrict record visibility based on the Branch field value.
3. **Script-Controlled ACL:** Use server-side scripts to evaluate user roles and grant or deny access.
4. **READ ACL:** Allow authorized users to view only EEE branch records.
5. **CREATE ACL:** Allow users with the required creation role to create records.
6. **WRITE ACL:** Allow users with the required write role to modify records.
7. **DELETE ACL:** Allow users with the required delete role to delete records.
8. **Administrator Access:** Allow administrators to access all records.
9. **Security Testing:** Impersonate users with different roles to verify access permissions.

Grouping of Ideas

- **User Management:** Create test users and assign roles such as bb1, bb2, bb3, and bb4.
- **Data Management:** Create the Institution Details table with student information and branch values ECE, EEE, and CSE.
- **Access Control:** Configure separate READ, CREATE, WRITE, and DELETE ACLs.
- **Testing and Validation:** Verify permissions by impersonating EEE users and administrators.

Step 3: Idea Prioritization

Priority	Idea	Reason for Selection
1	Script-Controlled READ ACL	Restricts record visibility based on user roles and branch conditions.
2	Role-Based Permissions	Separates viewing, creating, editing, and deleting permissions.
3	Field-Based Security	Ensures that access to records follows the configured branch condition.
4	Administrator Access	Enables administrators to manage all records.
5	Impersonation and Testing	Verifies that the configured ACLs enforce the intended permissions.

Selected Best Idea

Implement script-controlled ACLs in ServiceNow using user roles and the Branch field to secure student records. Configure separate READ, CREATE, WRITE, and DELETE permissions and validate the results through user impersonation.

Expected Outcome

The project demonstrates how ServiceNow ACLs can protect student records from unauthorized access. Users receive permissions according to their assigned roles and the configured security conditions, while administrators retain full access. The implementation improves data confidentiality, access management, and record security.

Define the Problem Statements

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PS-1

Problem Statement (PS-1)

I am (Customer): A student or faculty member using the ServiceNow institution records system.

I'm trying to: View student records that belong only to my authorized academic branch, particularly EEE.

But: I may be able to access records from other branches if proper access restrictions are not configured.

Because: The system needs field-based access control and role-based permissions to prevent unauthorized users from viewing sensitive student information.

Which makes me feel: Concerned about the privacy and security of student records.

PS-2

Problem Statement (PS-2)

I am (Customer): A ServiceNow administrator responsible for managing student record access.

I'm trying to: Control who can read, create, update, and delete institution records based on assigned roles.

But: Managing different permissions without properly configured Access Control Lists can lead to unauthorized record access or modification.

Because: Each operation requires appropriate ACL rules, user roles, and security conditions.

Which makes me feel: Concerned about data security, unauthorized changes, and the reliability of access management.

IDEATION PHASE – EMPATHIZE & DISCOVER

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Empathy Map Canvas

1. WHO is the user?

- ServiceNow administrator
- Students and faculty members who access Institution Details records
- Users with different roles and branch permissions

2. What does the user THINK & FEEL?

- “My personal and academic details should be secure.”
- “I should only access records that I am authorized to view.”
- “Unauthorized users should not be able to modify or delete records.”
- Concerned about privacy and data security.

3. What does the user SEE?

- Institution Details records containing student names, roll numbers, branch, email, and phone numbers.
- Users with different roles and access permissions.
- Access Control Rules (ACLs) used to control record access.

4. What does the user SAY & DO?

- Requests access to authorized institution records.
- Checks whether users can access records based on their roles.
- Configures READ, CREATE, WRITE, and DELETE ACLs.
- Tests permissions to ensure unauthorized access is restricted.

5. What does the user HEAR?

- Instructions from administrators about access permissions.
- Security requirements for protecting student information.
- Feedback about unauthorized access or incorrect permissions.
- Recommendations to apply role-based and field-based access control.

6. PAINS (Challenges)

- Unauthorized users may access sensitive student information.
- Incorrect ACL configuration may allow unwanted record changes.
- Managing separate permissions for different roles can be difficult.
- Security errors may affect data privacy and reliability.

7. GAINS (Needs and Benefits)

- Secure access to Institution Details records.
- EEE branch users can be restricted to EEE records when the ACL conditions are correctly configured.
- Separate permissions for READ, CREATE, WRITE, and DELETE operations.
- Administrators retain full access.
- Improved privacy, data protection, and access control.

Main Insight

Users need a secure ServiceNow system that restricts institution records according to roles and branch values, while allowing administrators full access.